EHR Project

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2023-11-28

```r
library(tidyverse)
library(lubridate)
library(survival)
library(epiR)
library(epiDisplay)
library(cmprsk)
library(ggplot2)
library(lme4)
```

```r
hosp <- read.csv("hospitalisation_2018-2020.csv")
pres <- read.csv("prescription_2018-2020.csv")
diab_h <- read.csv("diabetes_hosp_2018-2020.csv")
diab_l <- read.csv("diabetes_lti_2018-2020.csv")
pat <- read.csv("tab_patient.csv")
```

```r
## PATIENT tables -----------------------------------------------------

# Convert date columns to date format
pat$PAT_BIRTH_DATE <- date(pat$PAT_BIRTH_DATE)

# create placeholder death date for NAs to calculate age at end of study period
unique(pat$PAT_DEATH_DATE)   # 2020-12-31 is a death date so will set 2021-01-01 as placeholder
```

```
##   [1] ""           "2020-06-24" "2020-05-17" "2020-12-31" "2020-12-29"
##   [6] "2020-11-23" "2021-08-11" "2020-12-12" "2020-09-27" "2021-09-03"
##  [11] "2021-11-03" "2021-01-10" "2021-10-09" "2020-08-28" "2021-07-21"
##  [16] "2020-12-25" "2021-09-07" "2021-09-14" "2021-10-02" "2021-05-24"
##  [21] "2021-05-09" "2021-10-12" "2021-02-04" "2020-12-11" "2021-08-29"
##  [26] "2021-12-20" "2020-12-28" "2020-11-29" "2021-01-01" "2021-05-11"
##  [31] "2020-11-04" "2021-10-05" "2021-11-24" "2020-12-09" "2021-08-24"
##  [36] "2020-06-06" "2021-10-19" "2020-12-21" "2020-12-02" "2020-11-28"
##  [41] "2021-03-06" "2021-05-23" "2021-10-31" "2021-03-10" "2020-12-17"
##  [46] "2021-05-15" "2020-12-20" "2020-12-16" "2021-09-06" "2021-09-12"
##  [51] "2021-11-19" "2021-10-28" "2021-09-13" "2021-09-19" "2020-12-30"
##  [56] "2020-12-19" "2021-01-30" "2021-11-12" "2021-12-15" "2021-05-17"
##  [61] "2020-12-26" "2021-01-08" "2021-04-10" "2021-10-16" "2021-09-10"
##  [66] "2021-08-03" "2021-06-17" "2020-04-06" "2021-06-22" "2021-07-18"
##  [71] "2021-01-09" "2021-02-03" "2021-02-24" "2020-11-25" "2021-09-23"
##  [76] "2021-02-25" "2021-06-30" "2020-12-07" "2021-08-16" "2021-04-15"
##  [81] "2021-06-07" "2021-01-24" "2021-06-24" "2021-01-03" "2021-12-02"
##  [86] "2021-03-03" "2021-06-10" "2021-02-10" "2021-06-13" "2021-08-17"
##  [91] "2021-06-09" "2020-08-20" "2021-03-02" "2021-11-26" "2021-03-01"
##  [96] "2021-11-10" "2021-08-20" "2021-08-08" "2020-12-13" "2021-08-23"
```

```
## [101] "2021-10-03" "2021-12-01" "2021-10-08" "2021-07-15" "2021-04-21"
## [106] "2021-12-06" "2021-04-07" "2021-09-18" "2020-12-24" "2021-09-16"
## [111] "2021-06-12" "2021-11-06" "2021-07-28" "2020-12-22"
```

```
pat$PAT_DEATH_DATE <- date(ifelse(pat$PAT_DEATH_DATE == "",
                                  "2021-01-01", pat$PAT_DEATH_DATE))

dates <- pat$PAT_DEATH_DATE
dates[dates >= ymd("2021-01-01")] <- ymd("2021-01-01")
pat$PAT_DEATH_DATE <- dates

# Status indicator at end of study (0 = alive, 1 = death)
pat$STATUS <- ifelse(pat$PAT_DEATH_DATE == "", 0, 1)

# Calculate age at death or age at end of study period
pat$PAT_AGE <- as.double(difftime(pat$PAT_DEATH_DATE, pat$PAT_BIRTH_DATE,
                                  units = "weeks")) / 52

# Group by geographic area(Northeast, Northwest, Southeast, Southwest,
# Île-de-France, French territory islands (3-digits))
pat <- pat %>% add_column(PAT_DPT_GRP = NA, .after = "PAT_DPT_RES")

table(pat$PAT_DPT_RES)   # check unique values
```

```
##
##   0   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19
##   3  89  65  44  19  19 198  47  32  20  41  44  41 324 109  22  50  83  33  30
##  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39
##  17  65  85  12  50  73  87  84  52 135 102 199  22 263 213 132  15  81 224  26
##  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59
##  52  37 120  32 186  80  13  44   9 116  55  62  30  37 114  18 136 165  23 386
##  60  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79
## 129  37 217  80 102  36  68 160  98 268  28  69  70  67 126 340 188 229 229  57
##  80  81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  97  98  99 209
##  66  55  44 176  67 115  62  63  46  58  31 188 237 268 240 181 314   6  85  23
## 971 972 973 974 978 999
##   1   3   1   7   2  98
```

```
pat$PAT_DPT_GRP <- ifelse(pat$PAT_DPT_RES %in% c("2", "8", "10", "21", "25", "39",
                                                  "51", "52", "54", "55", "57",
                                                  "59", "62", "67", "68", "70",
                                                  "71", "88", "89", "90"), "NE",
       ifelse(pat$PAT_DPT_RES %in% c("14", "18", "22", "27", "28", "29", "35", "37",
                                     "41", "44", "45", "49", "50", "53", "56", "58",
                                     "60", "61", "72", "76", "80", "85"), "NW",
              ifelse(pat$PAT_DPT_RES %in% c("1", "3", "4", "5", "6", "7", "13", "15",
                                            "26", "30", "34", "38", "42", "43", "63",
                                            "66", "69", "73", "74", "83", "84"), "SE",
                     ifelse(pat$PAT_DPT_RES %in% c("9", "11", "12", "16", "17", "19",
                                                   "23", "24", "31", "32", "33",
                                                   "36", "40", "46", "47", "48",
                                                   "64", "65", "79", "81", "82",
                                                   "86", "87"), "SW",
                            ifelse(pat$PAT_DPT_RES %in% c("75", "77", "78", "91", "92",
                                                          "93", "94", "95"), "IF",
```

```r
                                    ifelse(pat$PAT_DPT_RES %in% c("0", "20", "97",
                                                                  "98", "99", "209",
                                                                  "971", "972", "973",
                                                                  "974", "978", "999"),
                                           "Other", "Unknown"))))))

# check which depts are still missing groupings
pat %>% filter(PAT_DPT_GRP == "Unknown") %>%
  distinct(PAT_DPT_RES)
```

```
## [1] PAT_DPT_RES
## <0 rows> (or 0-length row.names)
```

```r
# Diabetic patients -- convert date columns to date format
diab_h$HOSP_START_DATE <- date(diab_h$HOSP_START_DATE)
diab_h$HOSP_END_DATE <- date(diab_h$HOSP_END_DATE)

diab_l$LTI_START_DATE <- date(diab_l$LTI_START_DATE)
diab_l$LTI_END_DATE <- date(diab_l$LTI_END_DATE)

# check
class(diab_h$HOSP_START_DATE)
```

```
## [1] "Date"
```

```r
class(diab_h$HOSP_END_DATE)
```

```
## [1] "Date"
```

```r
class(diab_l$LTI_START_DATE)
```

```
## [1] "Date"
```

```r
class(diab_l$LTI_END_DATE)
```

```
## [1] "Date"
```

```r
## PRESCRIPTION table -------------------------------------------------

# Convert date columns to date format
pres$PRS_DATE <- date(pres$PRS_DATE)


## HOSPITALIZATION table ----------------------------------------------

# Convert date columns to date format
hosp$HOSP_START_DATE <- date(hosp$HOSP_START_DATE)
hosp$HOSP_END_DATE <- date(hosp$HOSP_END_DATE)

# check
class(hosp$HOSP_START_DATE)
```

```
## [1] "Date"
```

```r
class(hosp$HOSP_END_DATE)
```

```
## [1] "Date"
```

```r
# Calculate hospitalization duration (in days)
hosp <- hosp %>% add_column(HOSP_DUR = NA, .after = "HOSP_END_DATE")
hosp$HOSP_DUR <- as.numeric(difftime(hosp$HOSP_END_DATE, hosp$HOSP_START_DATE,
                                     units = "days"))

# Calculate number of hospitalizations
hosp_freq <- hosp %>% group_by(PAT_ID, HOSP_START_DATE) %>%
  summarise(freq = frequency(HOSP_START_DATE))
```

```
## `summarise()` has grouped output by 'PAT_ID'. You can override using the
## `.groups` argument.
```

```r
hosp_sum <- hosp_freq %>% group_by(PAT_ID) %>%
  summarise(NUM_HOSP = sum(freq))

hosp <- full_join(hosp, hosp_sum, by = "PAT_ID")

## COMBINING DATAFRAMES ---------------------------------------------

# Combine T2D patients from LTI and from
diab <- full_join(diab_h, diab_l, by = "PAT_ID")

## Clean up the dataframes ------------------------------------------
# Prescription table
pres <- pres %>%
  dplyr::select(-c("DRUG_CIP7.1","DRUG_BIG_PACK","DRUG_FRM_GAL","DRUG_LAB_ID","PRS_HP_COD","PRS_NAT"))

# Hospitalization table
hops <- hosp %>%
  dplyr::select(-c("ETA_NUM","RSA_NUM","HOSP_ENT_MODE","HOSP_ENT_ORIGIN","HOSP_EXIT_MODE","HOSP_EXIT_DES

# Patient table
pat <- pat %>%
  dplyr::select(-c("PAT_HEA_INSUR","PAT_LOC_IRIS"))

# Diabetic Patient Table
diab <- diab %>%
  dplyr::select(-c("ETA_NUM","RSA_NUM","HOSP_ENT_MODE","HOSP_ENT_ORIGIN","HOSP_EXIT_MODE","HOSP_EXIT_DES

# Find the patient information that only has a diagbetic diagnosis
diagnostic <- inner_join(pat, diab, by="PAT_ID")
dim(diagnostic)
```

```
## [1] 518  16
```

```r
# Join the prescription with patient data by PAT_ID
prescription <- full_join(diagnostic, pres, by = "PAT_ID")
```

```
## Warning in full_join(diagnostic, pres, by = "PAT_ID"): Detected an unexpected many-to-many relationsl
## i Row 1 of `x` matches multiple rows in `y`.
## i Row 781 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```r
prescription$PAT_ID <- as.character(prescription$PAT_ID)

class(prescription)
```

```
## [1] "data.frame"
```

```r
dim(prescription)
```

```
## [1] 41114    22
```

```r
# Make sure all the data we need are in the correct format
prescription$PAT_ID <- as.character(prescription$PAT_ID)
prescription$PAT_SEX_COD <- as.character(prescription$PAT_SEX_COD)
class(prescription$PAT_AGE)
```

```
## [1] "numeric"
```

```r
class(diagnostic$PAT_SEX_COD)
```

```
## [1] "integer"
```

```r
diagnostic$PAT_ID<- as.character(diagnostic$PAT_ID)
diagnostic$PAT_SEX_COD <- as.character(diagnostic$PAT_SEX_COD)
```

Logistic Regression

Logistic regression will then be performed with the explanatory variable being the four geographic areas, and the response variable being the binary outcome of the presence of at least one anti-diabetic drug prescription during the study period. An unadjusted model will be first conducted to investigate to understand an overall association between the geographic location of the patient and the presence of anti-diabetic drug prescription. A multivariate logistic regression will then be performed with age and sex as covariates to understand the impact of age and sex and confounders. The unadjusted and multivariate logistic models will then be compared to understand the impact of geographic location of diabetic patients on the prescription and intake of anti-diabetic drugs with and without the presence of confounders.

```r
#Recode hospitalization into binary variable
diagnostic$hosp <- diagnostic$HOSP_START_DATE
diagnostic$hosp <- ifelse(is.na(diagnostic$HOSP_START_DATE), 0, 1)

#Apply generalized linear model
g1 <- glm(hosp~PAT_DPT_GRP, data=diagnostic, family = "binomial")
summary(g1)
```

```
##
## Call:
## glm(formula = hosp ~ PAT_DPT_GRP, family = "binomial", data = diagnostic)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)       0.58779    0.21082   2.788   0.0053 **
## PAT_DPT_GRPNE     0.23639    0.31936   0.740   0.4592
## PAT_DPT_GRPNW    -0.15247    0.30717  -0.496   0.6196
## PAT_DPT_GRPOther  0.90029    0.46857   1.921   0.0547 .
## PAT_DPT_GRPSE     0.04218    0.27518   0.153   0.8782
## PAT_DPT_GRPSW     0.22796    0.32734   0.696   0.4862
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 658.50  on 517  degrees of freedom
## Residual deviance: 652.07  on 512  degrees of freedom
## AIC: 664.07
##
## Number of Fisher Scoring iterations: 4
```

```r
r1 <- exp(g1$coefficients)
c1 <- exp(confint(g1))
```

```
## Waiting for profiling to be done...
```

```r
cbind(OR=r1, c1)
```

```
##                          OR     2.5 %   97.5 %
## (Intercept)       1.8000000 1.1987137 2.747455
## PAT_DPT_GRPNE     1.2666667 0.6792808 2.383945
## PAT_DPT_GRPNW     0.8585859 0.4695029 1.569488
## PAT_DPT_GRPOther  2.4603175 1.0267765 6.597771
## PAT_DPT_GRPSE     1.0430839 0.6062008 1.786649
## PAT_DPT_GRPSW     1.2560386 0.6637676 2.404502
```

```r
#Multivariate logistic regression
g2 <- glm(hosp~PAT_DPT_GRP+PAT_AGE+PAT_SEX_COD, data=diagnostic, family="binomial")
summary(g2)
```

```
##
## Call:
## glm(formula = hosp ~ PAT_DPT_GRP + PAT_AGE + PAT_SEX_COD, family = "binomial",
##     data = diagnostic)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -5.88311    0.66730  -8.816   <2e-16 ***
## PAT_DPT_GRPNE      0.35838    0.39944   0.897    0.370
## PAT_DPT_GRPNW     -0.38534    0.38844  -0.992    0.321
## PAT_DPT_GRPOther   0.81660    0.56382   1.448    0.148
## PAT_DPT_GRPSE      0.17939    0.35326   0.508    0.612
## PAT_DPT_GRPSW      0.26669    0.41840   0.637    0.524
## PAT_AGE            0.10839    0.01015  10.682   <2e-16 ***
## PAT_SEX_COD2       0.02116    0.24393   0.087    0.931
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 658.50  on 517  degrees of freedom
## Residual deviance: 454.96  on 510  degrees of freedom
## AIC: 470.96
##
## Number of Fisher Scoring iterations: 5
```

```r
cbind(OR=exp(g2$coefficients),exp(confint(g2)))
```

```
## Waiting for profiling to be done...
```

```
##                           OR        2.5 %      97.5 %
## (Intercept)      0.002786102 0.0007114171 0.009781995
## PAT_DPT_GRPNE    1.431005318 0.6557154003 3.153040319
## PAT_DPT_GRPNW    0.680222422 0.3157703424 1.453571713
## PAT_DPT_GRPOther 2.262796028 0.7836468720 7.315960421
## PAT_DPT_GRPSE    1.196489135 0.5969277742 2.393011348
## PAT_DPT_GRPSW    1.305639990 0.5770840167 2.991137925
## PAT_AGE          1.114484796 1.0935192912 1.137985484
## PAT_SEX_COD2     1.021385267 0.6347662756 1.654877259
```

```
#Recode prescription into binary variable
prescription$pres <- prescription$DRUG_ATC_C07
prescription$pres <- ifelse(str_detect(prescription$DRUG_ATC_C07, "A10"), 1, 0)
prescription$pres <- ifelse(is.na(prescription$DRUG_NAME), 0, prescription$pres)

#Apply generalized linear model
g3 <- glm(pres~PAT_DPT_GRP, data=prescription, family = "binomial")
summary(g3)
```

```
##
## Call:
## glm(formula = pres ~ PAT_DPT_GRP, family = "binomial", data = prescription)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.00873    0.03203 -31.491  < 2e-16 ***
## PAT_DPT_GRPNE     0.14018    0.04434   3.161  0.00157 **
## PAT_DPT_GRPNW    -0.11115    0.04577  -2.428  0.01517 *
## PAT_DPT_GRPOther  0.11287    0.06474   1.744  0.08124 .
## PAT_DPT_GRPSE    -0.08245    0.04088  -2.017  0.04372 *
## PAT_DPT_GRPSW     0.01767    0.04641   0.381  0.70339
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 34037  on 29390  degrees of freedom
## Residual deviance: 33990  on 29385  degrees of freedom
##   (11723 observations deleted due to missingness)
## AIC: 34002
##
## Number of Fisher Scoring iterations: 4
```

```
cbind(OR=exp(g3$coefficients),exp(confint(g3)))
```

```
## Waiting for profiling to be done...
```

```
##                         OR     2.5 %    97.5 %
## (Intercept)      0.3646833 0.3423847 0.3881962
## PAT_DPT_GRPNE    1.1504804 1.0547545 1.2549994
## PAT_DPT_GRPNW    0.8948074 0.8180049 0.9787670
## PAT_DPT_GRPOther 1.1194833 0.9855006 1.2702345
## PAT_DPT_GRPSE    0.9208562 0.8500362 0.9977934
## PAT_DPT_GRPSW    1.0178266 0.9293011 1.1147233
```

```r
#Multivariate logistic regression
g4 <- glm(pres~PAT_DPT_GRP+PAT_AGE+PAT_SEX_COD, data=prescription, family="binomial")
summary(g4)
```

```
##
## Call:
## glm(formula = pres ~ PAT_DPT_GRP + PAT_AGE + PAT_SEX_COD, family = "binomial",
##     data = prescription)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.570221   0.079587  -7.165 7.80e-13 ***
## PAT_DPT_GRPNE     0.129131   0.044481   2.903 0.003695 **
## PAT_DPT_GRPNW    -0.105762   0.045835  -2.307 0.021031 *
## PAT_DPT_GRPOther  0.078380   0.065264   1.201 0.229761
## PAT_DPT_GRPSE    -0.076731   0.040927  -1.875 0.060817 .
## PAT_DPT_GRPSW     0.003646   0.046599   0.078 0.937630
## PAT_AGE          -0.005769   0.001041  -5.540 3.02e-08 ***
## PAT_SEX_COD2     -0.093847   0.027330  -3.434 0.000595 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 34037  on 29390  degrees of freedom
## Residual deviance: 33944  on 29383  degrees of freedom
##   (11723 observations deleted due to missingness)
## AIC: 33960
##
## Number of Fisher Scoring iterations: 4
```

```r
cbind(OR=exp(g4$coefficients),exp(confint(g4)))
```

```
## Waiting for profiling to be done...
```

```
##                         OR     2.5 %    97.5 %
## (Intercept)      0.5654005 0.4836144 0.6606875
## PAT_DPT_GRPNE    1.1378393 1.0428788 1.2415469
## PAT_DPT_GRPNW    0.8996391 0.8223160 0.9841802
## PAT_DPT_GRPOther 1.0815341 0.9511169 1.2284579
## PAT_DPT_GRPSE    0.9261392 0.8548394 1.0036052
## PAT_DPT_GRPSW    1.0036530 0.9160152 1.0996083
## PAT_AGE          0.9942472 0.9922218 0.9962806
## PAT_SEX_COD2     0.9104220 0.8628878 0.9604653
```

Linear Regression

A linear regression will be used to investigate the impact of geographic location of the diabetic patients on the secondary outcome, the number of hospitalizations. Potential confounders considered include age and sex.

```r
#Calculate the number of hospitalizations
hosp$PAT_ID <- as.character(hosp$PAT_ID)
hospitalization <- inner_join(diagnostic, hosp, by="PAT_ID")
```

```
## Warning in inner_join(diagnostic, hosp, by = "PAT_ID"): Detected an unexpected many-to-many relations
## i Row 1 of `x` matches multiple rows in `y`.
```

```
## i Row 3782 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```r
#unadjusted linear model
l1 <- lm(NUM_HOSP~PAT_DPT_GRP, data=hospitalization)
summary(l1)
```

```
##
## Call:
## lm(formula = NUM_HOSP ~ PAT_DPT_GRP, data = hospitalization)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -52.722 -14.946  -0.051  15.054  48.054
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        51.9455     0.3364  154.41   <2e-16 ***
## PAT_DPT_GRPNE      -5.8676     0.5042  -11.64   <2e-16 ***
## PAT_DPT_GRPNW     -11.8343     0.5188  -22.81   <2e-16 ***
## PAT_DPT_GRPOther   5.7766     0.5773   10.01   <2e-16 ***
## PAT_DPT_GRPSE      -4.1699     0.4431   -9.41   <2e-16 ***
## PAT_DPT_GRPSW      -8.8947     0.5162  -17.23   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.49 on 17258 degrees of freedom
## Multiple R-squared:  0.06215,    Adjusted R-squared:  0.06188
## F-statistic: 228.8 on 5 and 17258 DF,  p-value: < 2.2e-16
```

```r
cbind(Coefficient=l1$coefficient, confint(l1))
```

```
##                   Coefficient      2.5 %     97.5 %
## (Intercept)         51.945503  51.286093  52.604914
## PAT_DPT_GRPNE       -5.867581  -6.855820  -4.879343
## PAT_DPT_GRPNW      -11.834301 -12.851146 -10.817456
## PAT_DPT_GRPOther     5.776558   4.645057   6.908060
## PAT_DPT_GRPSE       -4.169890  -5.038490  -3.301290
## PAT_DPT_GRPSW       -8.894676  -9.906518  -7.882835
```

```r
#adjusted linear model
l2 <- lm(NUM_HOSP~PAT_DPT_GRP+PAT_AGE+PAT_SEX_COD, data=hospitalization)
summary(l2)
```

```
##
## Call:
## lm(formula = NUM_HOSP ~ PAT_DPT_GRP + PAT_AGE + PAT_SEX_COD,
##     data = hospitalization)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -53.026 -14.862   0.159  13.170  45.237
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)       39.8977     0.8152  48.946  < 2e-16 ***
## PAT_DPT_GRPNE     -5.5008     0.4992 -11.019  < 2e-16 ***
## PAT_DPT_GRPNW    -11.8749     0.5132 -23.138  < 2e-16 ***
## PAT_DPT_GRPOther   4.9788     0.5761   8.642  < 2e-16 ***
## PAT_DPT_GRPSE     -3.5574     0.4398  -8.089 6.44e-16 ***
## PAT_DPT_GRPSW     -8.6497     0.5126 -16.873  < 2e-16 ***
## PAT_AGE            0.1524     0.0107  14.246  < 2e-16 ***
## PAT_SEX_COD2       4.2647     0.3019  14.127  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.29 on 17256 degrees of freedom
## Multiple R-squared:  0.08229,    Adjusted R-squared:  0.08191
## F-statistic:   221 on 7 and 17256 DF,  p-value: < 2.2e-16
```

```
cbind(Coefficient=l2$coefficient, confint(l2))
```

```
##                  Coefficient       2.5 %     97.5 %
## (Intercept)       39.8977308  38.2999630  41.495499
## PAT_DPT_GRPNE     -5.5007777  -6.4792770  -4.522278
## PAT_DPT_GRPNW    -11.8749448 -12.8808991 -10.868991
## PAT_DPT_GRPOther   4.9787585   3.8494934   6.108024
## PAT_DPT_GRPSE     -3.5574269  -4.4194999  -2.695354
## PAT_DPT_GRPSW     -8.6497484  -9.6545575  -7.644939
## PAT_AGE            0.1524243   0.1314526   0.173396
## PAT_SEX_COD2       4.2647072   3.6729947   4.856420
```

Mixed Models

Mixed model will be performed to understand the association between the number of hospitalization and the duration of each hospitalization stratified by each geographic area of the diabetic patients. Potential confounders include age, sex, and number of prescriptions.

```
hospitalization$PAT_ID <- as.character(hospitalization$PAT_ID)

# Run the mixed model for different geographic departments only
# only random intercept
m1 <- lmer(HOSP_DUR~NUM_HOSP+(1|PAT_ID), hospitalization, REML=0)
m1
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID)
##    Data: hospitalization
##       AIC       BIC    logLik  deviance  df.resid
## 140180.89 140211.91 -70086.44 140172.89     17260
## Random effects:
##  Groups   Name        Std.Dev.
##  PAT_ID   (Intercept)  3.336
##  Residual             13.858
## Number of obs: 17264, groups:  PAT_ID, 414
## Fixed Effects:
## (Intercept)     NUM_HOSP
##    6.779828    -0.002116
```

```
summary(m1)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
```

```
## Formula: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID)
##    Data: hospitalization
##
##      AIC      BIC   logLik deviance df.resid
## 140180.9 140211.9 -70086.4 140172.9    17260
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4232 -0.3823 -0.1721  0.1688 28.5605
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  PAT_ID   (Intercept)  11.13    3.336
##  Residual             192.05   13.858
## Number of obs: 17264, groups:  PAT_ID, 414
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  6.779828   0.462104  14.672
## NUM_HOSP    -0.002116   0.010862  -0.195
##
## Correlation of Fixed Effects:
##          (Intr)
## NUM_HOSP -0.880
```

```
# random intercept + random slope
m2 <- lmer(HOSP_DUR~NUM_HOSP+(NUM_HOSP|PAT_ID), hospitalization, REML=0)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 1.58127 (tol = 0.002, component 1)

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly unide
##  - Rescale variables?
```

```
m2
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ NUM_HOSP + (NUM_HOSP | PAT_ID)
##    Data: hospitalization
##       AIC       BIC    logLik  deviance  df.resid
## 140068.81 140115.35 -70028.41 140056.81     17258
## Random effects:
##  Groups   Name        Std.Dev. Corr
##  PAT_ID   (Intercept)  9.5511
##           NUM_HOSP     0.1771  -1.00
##  Residual             13.7749
## Number of obs: 17264, groups:  PAT_ID, 414
## Fixed Effects:
## (Intercept)     NUM_HOSP
##    6.744435    -0.001317
## optimizer (nloptwrap) convergence code: 0 (OK) ; 0 optimizer warnings; 2 lme4 warnings
```

```
summary(m2)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ NUM_HOSP + (NUM_HOSP | PAT_ID)
##    Data: hospitalization
```

```
##
##      AIC      BIC   logLik deviance df.resid
## 140068.8 140115.4 -70028.4 140056.8    17258
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1825 -0.3882 -0.1704  0.1705 28.9025
##
## Random effects:
##  Groups   Name        Variance  Std.Dev. Corr
##  PAT_ID   (Intercept)  91.22436  9.5511
##           NUM_HOSP      0.03136  0.1771  -1.00
##  Residual             189.74661 13.7749
## Number of obs: 17264, groups:  PAT_ID, 414
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  6.744435   0.601152  11.219
## NUM_HOSP    -0.001317   0.012208  -0.108
##
## Correlation of Fixed Effects:
##          (Intr)
## NUM_HOSP -0.952
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 1.58127 (tol = 0.002, component 1)
## Model is nearly unidentifiable: very large eigenvalue
##  - Rescale variables?
```

```
# Run the mixed model for the duration of hospitalization that gives random intercept and random slope
m3 <- lmer(HOSP_DUR~NUM_HOSP+(1|PAT_ID)+(0+NUM_HOSP|PAT_ID), hospitalization, REML=0)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
m3
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID) + (0 + NUM_HOSP | PAT_ID)
##    Data: hospitalization
##       AIC       BIC    logLik  deviance  df.resid
## 140182.89 140221.67 -70086.44 140172.89     17259
## Random effects:
##  Groups   Name        Std.Dev.
##  PAT_ID   (Intercept)  3.336
##  PAT_ID.1 NUM_HOSP     0.000
##  Residual             13.858
## Number of obs: 17264, groups:  PAT_ID, 414
## Fixed Effects:
## (Intercept)     NUM_HOSP
##    6.779828    -0.002116
## optimizer (nloptwrap) convergence code: 0 (OK) ; 0 optimizer warnings; 1 lme4 warnings
```

```
summary(m3)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID) + (0 + NUM_HOSP | PAT_ID)
##    Data: hospitalization
```

```
##
##      AIC      BIC   logLik deviance df.resid
## 140182.9 140221.7 -70086.4 140172.9    17259
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4232 -0.3823 -0.1721  0.1688 28.5605
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  PAT_ID   (Intercept)  11.13    3.336
##  PAT_ID.1 NUM_HOSP      0.00    0.000
##  Residual             192.05   13.858
## Number of obs: 17264, groups:  PAT_ID, 414
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  6.779828   0.462104  14.672
## NUM_HOSP    -0.002116   0.010862  -0.195
##
## Correlation of Fixed Effects:
##          (Intr)
## NUM_HOSP -0.880
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# Comparing the 3 models to decide the best model
anova(m1,m2)
```

```
## Data: hospitalization
## Models:
## m1: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID)
## m2: HOSP_DUR ~ NUM_HOSP + (NUM_HOSP | PAT_ID)
##    npar    AIC    BIC logLik deviance  Chisq Df Pr(>Chisq)
## m1    4 140181 140212 -70086   140173
## m2    6 140069 140115 -70028   140057 116.08  2  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m1,m3)
```

```
## Data: hospitalization
## Models:
## m1: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID)
## m3: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID) + (0 + NUM_HOSP | PAT_ID)
##    npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## m1    4 140181 140212 -70086   140173
## m3    5 140183 140222 -70086   140173     0  1          1
```

```
anova(m2,m3)
```

```
## Data: hospitalization
## Models:
## m3: HOSP_DUR ~ NUM_HOSP + (1 | PAT_ID) + (0 + NUM_HOSP | PAT_ID)
## m2: HOSP_DUR ~ NUM_HOSP + (NUM_HOSP | PAT_ID)
##    npar    AIC    BIC logLik deviance  Chisq Df Pr(>Chisq)
```

```
## m3    5 140183 140222 -70086   140173
## m2    6 140069 140115 -70028   140057 116.08  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# From the result, we don't see lots of differences among the 3 models. By BIC, m2 has the lowest BIC a
```

Sensitivity Analysis

```
# Adding the geographic departments to the equation to see if there is any change in the results.
m4 <- lmer(HOSP_DUR~PAT_DPT_GRP+NUM_HOSP+(NUM_HOSP|PAT_ID), hospitalization, REML=0)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 3.30871 (tol = 0.002, component 1)

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly unide
##  - Rescale variables?
```

```
m4
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ PAT_DPT_GRP + NUM_HOSP + (NUM_HOSP | PAT_ID)
##    Data: hospitalization
##       AIC       BIC    logLik  deviance  df.resid
## 140070.47 140155.79 -70024.23 140048.47     17253
## Random effects:
##  Groups   Name        Std.Dev. Corr
##  PAT_ID   (Intercept)  9.3509
##           NUM_HOSP     0.1781  -1.00
##  Residual             13.7829
## Number of obs: 17264, groups:  PAT_ID, 414
## Fixed Effects:
##      (Intercept)     PAT_DPT_GRPNE     PAT_DPT_GRPNW  PAT_DPT_GRPOther
##        6.6309032        -0.6455132         1.0008925        -0.3878075
##    PAT_DPT_GRPSE     PAT_DPT_GRPSW          NUM_HOSP
##       -0.0285302        -0.2996561         0.0004664
## optimizer (nloptwrap) convergence code: 0 (OK) ; 0 optimizer warnings; 2 lme4 warnings
```

```
summary(m4)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: HOSP_DUR ~ PAT_DPT_GRP + NUM_HOSP + (NUM_HOSP | PAT_ID)
##    Data: hospitalization
##
##      AIC      BIC   logLik deviance df.resid
## 140070.5 140155.8 -70024.2 140048.5    17253
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1269 -0.3884 -0.1708  0.1719 28.9049
##
## Random effects:
##  Groups   Name        Variance  Std.Dev. Corr
##  PAT_ID   (Intercept)  87.44012  9.3509
##           NUM_HOSP      0.03173  0.1781  -1.00
##  Residual             189.96905 13.7829
```

```
## Number of obs: 17264, groups:  PAT_ID, 414
##
## Fixed effects:
##                     Estimate Std. Error t value
## (Intercept)        6.6309032  0.7124112   9.308
## PAT_DPT_GRPNE     -0.6455132  0.6243232  -1.034
## PAT_DPT_GRPNW      1.0008925  0.5858339   1.708
## PAT_DPT_GRPOther -0.3878075  0.7799036  -0.497
## PAT_DPT_GRPSE     -0.0285302  0.5513924  -0.052
## PAT_DPT_GRPSW     -0.2996561  0.6142272  -0.488
## NUM_HOSP           0.0004664  0.0120524   0.039
##
## Correlation of Fixed Effects:
##               (Intr) PAT_DPT_GRPNE PAT_DPT_GRPNW PAT_DPT_GRPO PAT_DPT_GRPSE
## PAT_DPT_GRPNE -0.413
## PAT_DPT_GRPNW -0.482  0.518
## PAT_DPT_GRPO  -0.323  0.390         0.414
## PAT_DPT_GRPSE -0.483  0.551         0.587         0.441
## PAT_DPT_GRPSW -0.443  0.494         0.528         0.396        0.560
## NUM_HOSP      -0.791 -0.017         0.034        -0.023        0.000
##               PAT_DPT_GRPSW
## PAT_DPT_GRPNE
## PAT_DPT_GRPNW
## PAT_DPT_GRPO
## PAT_DPT_GRPSE
## PAT_DPT_GRPSW
## NUM_HOSP       0.011
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 3.30871 (tol = 0.002, component 1)
## Model is nearly unidentifiable: very large eigenvalue
##  - Rescale variables?
```

```
# With all potential confounders
m5 <- lmer(HOSP_DUR~PAT_DPT_GRP+NUM_HOSP+PAT_AGE+PAT_SEX_COD++(NUM_HOSP|PAT_ID), hospitalization, REML=
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to conve

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly unide
##  - Rescale variables?
```

```
m5
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## HOSP_DUR ~ PAT_DPT_GRP + NUM_HOSP + PAT_AGE + PAT_SEX_COD + +(NUM_HOSP |
##     PAT_ID)
##    Data: hospitalization
##       AIC       BIC    logLik  deviance  df.resid
## 140064.06 140164.89 -70019.03 140038.06     17251
## Random effects:
##  Groups   Name        Std.Dev. Corr
##  PAT_ID   (Intercept)  9.2798
##           NUM_HOSP     0.1793  -1.00
##  Residual             13.7819
## Number of obs: 17264, groups:  PAT_ID, 414
## Fixed Effects:
```

```
##      (Intercept)     PAT_DPT_GRPNE     PAT_DPT_GRPNW  PAT_DPT_GRPOther
##        6.1775793        -0.5829416         0.9716814        -0.3833098
##    PAT_DPT_GRPSE     PAT_DPT_GRPSW          NUM_HOSP           PAT_AGE
##        0.1057863        -0.3300205        -0.0001831        -0.0007958
##     PAT_SEX_COD2
##        1.1493966
## optimizer (nloptwrap) convergence code: 0 (OK) ; 0 optimizer warnings; 2 lme4 warnings
```

```r
summary(m5)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## HOSP_DUR ~ PAT_DPT_GRP + NUM_HOSP + PAT_AGE + PAT_SEX_COD + +(NUM_HOSP |
##     PAT_ID)
##    Data: hospitalization
##
##      AIC      BIC   logLik deviance df.resid
## 140064.1 140164.9 -70019.0 140038.1    17251
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1203 -0.3832 -0.1681  0.1715 28.9027
##
## Random effects:
##  Groups   Name        Variance  Std.Dev. Corr
##  PAT_ID   (Intercept)  86.11543  9.2798
##           NUM_HOSP      0.03214  0.1793  -1.00
##  Residual             189.94123 13.7819
## Number of obs: 17264, groups:  PAT_ID, 414
##
## Fixed effects:
##                    Estimate Std. Error t value
## (Intercept)       6.1775793  1.0802012   5.719
## PAT_DPT_GRPNE    -0.5829416  0.6232574  -0.935
## PAT_DPT_GRPNW     0.9716814  0.5800461   1.675
## PAT_DPT_GRPOther -0.3833098  0.7722032  -0.496
## PAT_DPT_GRPSE     0.1057863  0.5528129   0.191
## PAT_DPT_GRPSW    -0.3300205  0.6050008  -0.545
## NUM_HOSP         -0.0001831  0.0122067  -0.015
## PAT_AGE          -0.0007958  0.0127747  -0.062
## PAT_SEX_COD2      1.1493966  0.3516348   3.269
##
## Correlation of Fixed Effects:
##               (Intr) PAT_DPT_GRPNE PAT_DPT_GRPNW PAT_DPT_GRPO PAT_DPT_GRPSE
## PAT_DPT_GRPNE -0.298
## PAT_DPT_GRPNW -0.281  0.516
## PAT_DPT_GRPO  -0.248  0.391         0.416
## PAT_DPT_GRPSE -0.382  0.549         0.579         0.442
## PAT_DPT_GRPSW -0.310  0.497         0.534         0.403        0.561
## NUM_HOSP      -0.409 -0.020         0.031        -0.034       -0.011
## PAT_AGE       -0.742  0.031        -0.037         0.052        0.076
## PAT_SEX_COD   -0.215  0.035        -0.032        -0.004        0.087
##               PAT_DPT_GRPSW NUM_HO PAT_AG
## PAT_DPT_GRPNE
## PAT_DPT_GRPNW
```

```
## PAT_DPT_GRPO
## PAT_DPT_GRPSE
## PAT_DPT_GRPSW
## NUM_HOSP       0.001
## PAT_AGE        0.031        -0.138
## PAT_SEX_COD   -0.014        -0.021  0.097
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.119764 (tol = 0.002, component 1)
## Model is nearly unidentifiable: very large eigenvalue
##  - Rescale variables?
```